# Maths 2(Revision)

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Q. find the LCM of two numbers.

↳ $a \times b = LCM \times HCF$.

↳ $LCM = \frac{a \times b}{HCF}$

HCF ↳ (a, b)

$a = 15 \rightarrow 1, 3, 5, 15$

$b = 35 \rightarrow 1, 5, 7, 35$

H.C.F = 5.

15, 35

1 — 15

1, 2, 3, 4, 5, - 15

$\frac{15}{1}, \frac{35}{1}$

CF = 1, 5

1 → min(a, b);

min returns minimum number

```
int hcf = 1;
for (int i = 1 ; i <= Math.min(a, b); i++){
    if (a % i == 0 && b % i == 0) hcf = i;
```

```
for(int i=1 ; i<=
        if(a%i==0 && b%i==0) hcf=i;
    }
Sout(hcf);
```

Q: Prime factorization of a number.

Sol:- $28 \rightarrow 2 \times 2 \times 7$

$23 \rightarrow 23$

$16 \rightarrow 2 \times 2 \times 2 \times 2$

int num = 16;

2 ⟶ 16

⟶

2, 3, 4, 5 ---- 16

$\frac{16}{2} = \frac{8}{2} = \frac{4}{2} = \frac{2}{2} = 1$

| 2 | 2 | 2 | 2 |
|---|---|---|---|

```
int n=num;
for(int i=2 ; i<=num; i++){
while(n%i==0){
    Sout(i);
    n=n/i;
}
}
```

$\frac{16}{2}$

$\Rightarrow 2$

23

i=3

$\boxed{2 \quad 2 \quad 2 \quad 2}$ $\Rightarrow 2$

$25 \rightarrow 5 \times 5$

$10 = 5 \times 2$

$\times 3$

$\times 4 \rightarrow 2 \times 2$

$1 \text{ — } n$

$1 \text{ — } \sqrt{n}$

$1, 2, \text{----} 36$

$36 \rightarrow (1, 36)$

$\hookrightarrow$ 2 unique factors

$\frac{36}{1} = 36 \quad (2, 18)$

$1 \rightarrow 36$

$1 \rightarrow 6$

$\frac{36}{2} = 18 \rightarrow (3, 12)$

$\rightarrow (4, 9) \rightarrow 2$

$\boxed{(6, 6)} \rightarrow 1$

$\rightarrow (9, 4)$

$\sqrt{36} = 6$

$4^2 = 16$

(4, 4)

(12, 3)

$4^2 = 16$

$5^2 = 25$

$\sqrt{24}$

$n = 24$

int a = 5/2

int a = $\sqrt{24}$

a = 2  4.9

$\boxed{a = 4}$

a = 4.9

$\boxed{a = 4}$

1, 24

2, 12

3, 8

(4, 6)

(6, 4)

```
int num = n;
for(int i = 2; i <= √n ; i++){
    while (num % i == 0){
        Sout(i);
        num = num/i;
    }
    if (num > 1) Sout(num);
```

$n = 28$ $\sqrt{n} = 5$

$28 \rightarrow 2, 2, 7$

2, 3, 4, 5

$\frac{28}{2} = 14 = \frac{7}{2}$ ✗ $\boxed{num = 7}$ → 8

2 2

~~(1) 28~~

(2, 14)

(4, 7)

~~(7, 4)~~

(14, 2)

(28, 1)

~~num = 7~~

$\frac{14}{2} = \frac{7}{2}$ ✗

2, 3, 4, 5

2 2

3, 4, 5

$i < = \sqrt{n}$

Sq. both sides

$i^2 < = n$

$i * i <= n$